

Swesan Pathmanathan

Canadian Citizen

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EDUCATION

McMaster University

Bachelor of Engineering, Software Engineering (Co-op)

Hamilton, ON

Sept 2021 – Apr 2026

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, Java, Bash

Hardware & Systems: Hardware Architecture, Hardware Troubleshooting, Platform Integration, System Design, Datacenter Systems, ASIC, Compute

System Management: Linux, Windows, BMC, BIOS/UEFI, Out-of-Band Management, Firmware, System Management

Validation & Testing: Test Planning, Hardware Validation, Operational Validation, Test Automation, System Verification

Infrastructure: Docker, Hardware Lifecycle, Process Development, Cross-Functional Collaboration

Databases: PostgreSQL, SQL, Database Design

Developer Tools: Git, Python Scripting, Test Frameworks

WORK EXPERIENCE

Advanced Micro Devices (AMD) | Python, C/C++, Linux, Windows, Hardware Architecture, Platform Integration, Test Planning, Operational Validation, BMC, BIOS/UEFI, Out-of-Band Management, System Design, Firmware, Docker

Markham, ON

System Integration & Validation Engineer

May 2024 – Aug 2025

- Led hardware platform readiness and test execution for new MI300X, MI325X, and MI350X datacenter GPUs, orchestrating cross-functional alignment with infrastructure and validation teams to integrate new platforms into operational environments; sustained **100+ validation runs weekly** to verify hardware quality and readiness for datacenter deployment.
- Developed and documented hardware platform bring-up procedures for **Linux** and **Windows** systems, including **BMC** and **BIOS/UEFI** configuration, validated baselines, and **Docker** workload staging; mentored 3+ team members on hardware integration best practices, reducing new-platform onboarding time and improving distributed team consistency.
- Set up **KVM/QEMU** on **MI325X CentOS** hosts, wrote setup docs the team reused for running workloads, validated **MI325X KVM** guests under production-style workloads, and triaged failures found during those runs.
- Diagnosed **BMC** and **BIOS/UEFI** failures across hardware platforms, resolved complex firmware and boot issues, and documented standardized **hardware troubleshooting procedures** adopted across the team on **20+ platform bring-ups**; improved onboarding efficiency and reduced ambiguity for new hardware integration cycles.
- Developed **Python** and **Bash**-based **test automation tooling** with internal launch frameworks and version-controlled procedures; influenced adoption across distributed validation teams, reducing manual test setup from **about 20 minutes to under 5 minutes** per cycle and improving consistency for new hardware platform bring-up.
- Triaged hardware and software integration failures across firmware, drivers, and platform code; developed and standardized reproducible **test procedures** and root-cause analysis documentation that influenced team practices, **reducing duplicate defect reruns by about 30%** and accelerating resolution cycles for new platform validation.
- Set up **virtualization environments** on datacenter GPU platforms using **KVM/QEMU**, validated guest workload execution under production-style loads, and documented setup procedures for team reuse across multiple platform generations.
- Validated datacenter rack-level **out-of-band management** (BMC, remote power/boot, inventory services) for new hardware platforms alongside operational validation tests; coordinated with distributed teams to improve platform operational readiness and keep lab infrastructure reliable for recurring integration campaigns.

PROJECTS

SufSort Sorting System | C, Bash, Linux

Nov 2025

- Wrote two sorting programs in **C** to compare how different algorithms behave on large data files of up to several hundred megabytes.

- Used **Bash** scripts on **Linux** to generate input data, run batch tests, and check sorted output, catching logic errors early in development.
- Measured runtime behavior across multiple input sizes and used test logs to compare sorting throughput, making algorithm tradeoffs visible for larger file workloads.

Ontario Hospital Association Database Design | SQL

Sept 2025

- Designed a relational schema for a hospital information system modeling patients, records, employees, and clinical workflows to support consistent operational queries.
- Authored SQL scripts implementing primary keys, foreign keys, and referential integrity constraints to ensure consistency across patient and record data.
- Developed a full database creation and deployment script with normalized relational schemas for reproducible SQL-based rollout.

ALU Design with Sequential Circuits | Verilog, Quartus

Sept 2025

- Structured a synthesizable ALU datapath from small **Verilog** blocks and checked behavior block-by-block before full-chip stitching.
- Coded deterministic **Verilog testbench** stimulus plus self-checks, exercised ALU arithmetic and logic modes over a scripted input matrix, and used **Quartus** waveform debug until every planned functional regression passed.

Linux Log Normalization and Token Frequency Analysis | Python, NumPy, pandas

Jan 2026

- Developed a configurable text normalization pipeline in **Python** for processing raw **Linux logs**, supporting tokenization, stemming, lemmatization, and digit filtering.
- Performed token frequency computation and statistical aggregation using **NumPy** and **pandas** to analyze distributional patterns in **Linux logs**.

Linux Log Corpus Analysis | Python, scikit-learn, NumPy

Feb 2026

- Built a simple analysis pipeline in **Python** to read and process tens of thousands of labeled Linux system logs.
- Used **NumPy** and **scikit-learn** to find which log terms best separate different log categories, improving signal-to-noise for downstream analysis.

Pneumonia Detection CNN | Python, TensorFlow/Keras, NumPy, pandas, scikit-learn

Jan 2026

- Built a convolutional neural network classifier in **Python** using **TensorFlow/Keras** to detect pneumonia from chest X-ray images.
- Applied data preprocessing, dataset splitting, and augmentation using **NumPy** and **pandas** to improve model generalization on imbalanced clinical datasets.
- Evaluated model performance using **accuracy**, **precision**, **recall**, confusion matrices, and visualization tools to guide architecture and hyperparameter tuning.

Terrain Generation System | Java, Maven, Git, JTS

2024

- Developed a software tool in **Java** for generating and visualizing 2D terrain meshes for simulation and visualization workflows.
- Modeled vertices, edges, and polygons using the **JTS** geometry library to represent terrain features and topology.
- Implemented a command-line interface to configure generation parameters, run batch experiments, and visualize output meshes.

Breast Cancer Prediction Neural Network | Python, PyTorch, NumPy, scikit-learn

Sept 2025 – Dec 2025

- Built a feedforward neural network with **PyTorch**, using **NumPy** and **scikit-learn** preprocessing and stratified splits, to classify malignant and benign breast cancer samples from tabular clinical datasets.
- Applied regularization techniques including **dropout** and **L1/L2** penalties to improve model stability and reduce overfitting.

RELEVANT COURSES

Software Testing, Computer Performance Analysis, Computer Architecture, Digital Systems Interfacing, Database Systems

HOBBIES & INTERESTS

Gaming, Hardware/Software Tinkering, Gym.